

Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs

1. [AJC/Prelims 2014/P1/14]

A car is making a turn at speed v . The radius of the turn is r and the centripetal force on the car is F . If the car rounds the same curve at speed $2v$, the required centripetal force is

- A $\frac{1}{2}F$ B F C $2F$ D $4F$

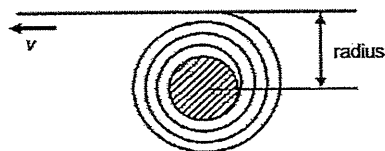
2. [AJC/Prelims 2014/P1/11]

A body of mass m moves in a horizontal circle of radius r at constant speed v for one complete revolution. Which statement is incorrect?

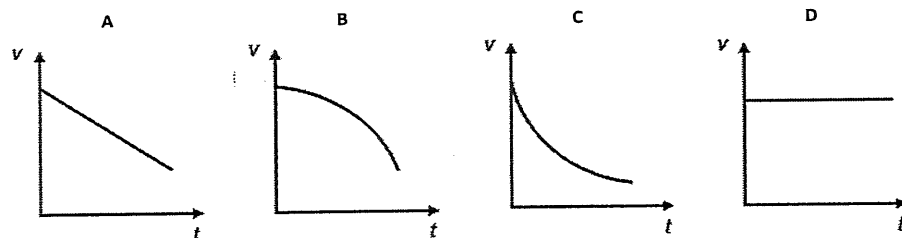
- A The angular velocity of body is directed perpendicular to the plane of circular motion.
 B The work done by the centripetal force is $2\pi mv^2$.
 C The change in linear momentum of the body is zero.
 D The total energy of the body is constant.

3. [AJC/Prelims 2014/P1/12]

A straight length of tape unwinds from a roll rotating about a fixed axis with constant angular velocity, the radius of the roll decreasing at a steady rate.



Which graph correctly shows how the variation of speed v at which the tape moves away from the roll with time t ?



4. [CJC/Prelims 2014/P1/11]

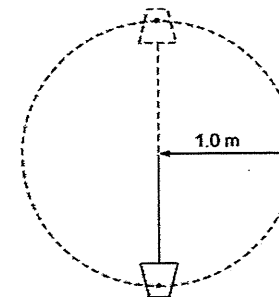
The clock on Big Ben has a minute hand of length 4.3 m and an hour hand of length 2.7 m.

What is the ratio of the centripetal acceleration of a point on the tip of the minute hand to a point on the tip of the hour hand?

- A 1.6 B 150 C 230 D 5700

5. [DHS/Prelims 2014/P1/11]

A light pail containing water is attached to a light rope and swung around in a vertical circle. The radius of the circular motion is 1.0 m. The water just manages to stay in the pail at the highest point of motion.

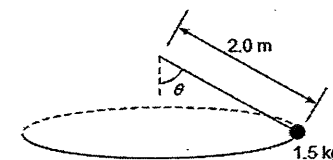


What is the linear speed of the pail at the lowest point of motion?

- A 3.13 m s^{-1} B 5.42 m s^{-1} C 7.00 m s^{-1} D 7.87 m s^{-1}

6. [DHS/Prelims 2014/P1/12]

A ball of mass 1.5 kg is attached to a light string of length 2.0 m and made to rotate in a horizontal circle.



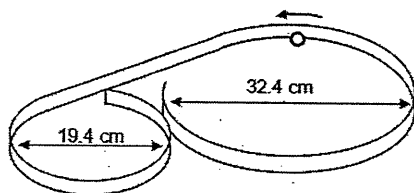
The string snaps when the linear speed of the ball reaches 12.0 m s^{-1} .

What is the maximum tension the string is able to withstand?

- A 18.9 N B 110 N C 217 N D 277 N

7. [HCI/Prelims 2014/P1/11]

A small sphere is travelling horizontally around the circumference of the bigger circular loop in the figure below with an angular velocity of 63.0 rad s^{-1} .



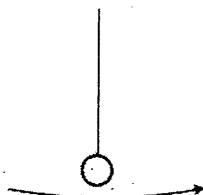
The sphere then moves into the smaller loop and continues to move along its circumference.

What will be the angular velocity of the sphere when it is moving in the smaller loop?

- A 37.7 rad s^{-1} B 63.0 rad s^{-1} C 105 rad s^{-1} D 126 rad s^{-1}

8. [HCI/Prelims 2014/P1/12]

A pendulum bob of mass 2.50 kg is supported by a string so that the radius of its path is 0.800 m . It is moving with velocity 0.475 m s^{-1} horizontally at the centre of its motion when the string is vertical.

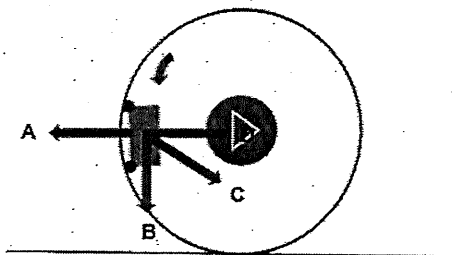


What is the tension in the string at this instant?

- A 0.705 N B 23.8 N C 25.2 N D 29.0 N

9. [IJC/Prelims 2014/P1/11]

The roller coaster car having just cleared the top of the loop is moving down a frictionless track at the moment shown.



Which direction A, B, C or D is the net force acting on the car?

10. [IJC/Prelims 2014/P1/12]

During the spin cycle of a washing machine, the clothes stick to the outer wall of the barrel as it spins at a rate of 1800 revolutions per minute about the vertical axis. The radius of the barrel is 36 cm .

What is the magnitude and direction of centripetal acceleration of the clothes which are located on the wall of the barrel?

	magnitude	direction
A	$46.0 \times 10^6 \text{ m s}^{-2}$	towards the centre of the barrel
B	$46.0 \times 10^6 \text{ m s}^{-2}$	away from the centre of the barrel
C	$12.8 \times 10^3 \text{ m s}^{-2}$	towards the centre of the barrel
D	$12.8 \times 10^3 \text{ m s}^{-2}$	away from the centre of the barrel

11. [IJC/Prelims 2014/P1/18]

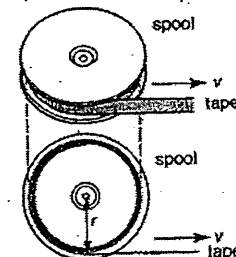
The maximum safe speed of a car rounding an unbanked corner is 16 m s^{-1} when the road is dry. The maximum frictional force between the road surface and wheels of the car is halved when the road is wet.

What is the maximum safe speed for the car to round the corner when the road is wet?

- A 4.0 m s^{-1} B 6.0 m s^{-1} C 8.0 m s^{-1} D 11 m s^{-1}

12. [IJC/Prelims 2014/P1/19]

In a tape cassette, the tape leaves one spool at a constant speed v and at a variable distance r from the centre.



Which statement is true?

- A The angular velocity decreases as radius r decreases.
 B The angular velocity increases as radius r decreases.
 C The angular velocity is directly proportional to the speed v .
 D The angular velocity is directly proportional to the speed v^2 .

13. [MJC/Prelims 2014/P1/11]

A rock attached to a string swings in a vertical circle. At the highest point,

- A no forces act on the rock.
 B only one force acts on the rock.
 C two forces act on the rock and their resultant is zero.
 D two forces act on the rock and their resultant is not zero.

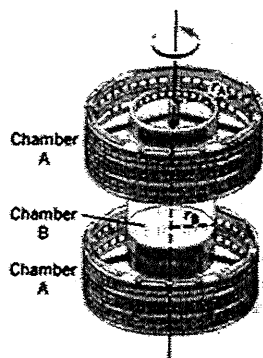
14. [MJC/Prelims 2014/P1/12]

A cyclist completes a circular path of circumference 343 m in 22 s on a rough horizontal surface. Assuming she cycles at constant speed, what is the angle made by her with the vertical?

- A 8.23° B 24.4° C 65.6° D 81.8°

15. [NJC/Prelims 2014/P1/12]

To create artificial gravity, the space station as shown is rotating at a rate of 1.00 revolutions per minute.



The radii of the cylindrical shaped chamber have the ratio of $r_A/r_B = 4.0$. Each chamber A simulates acceleration due to gravity of 10.0 m s^{-2} .

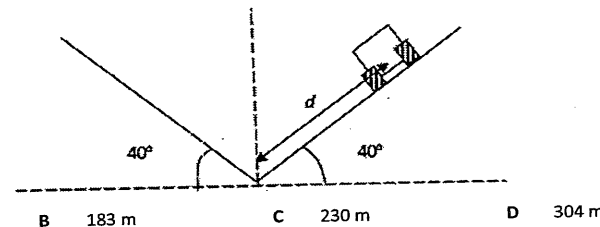
What is the tangential velocity of chamber B and its acceleration due to gravity that is simulated in it?

	tangential velocity of chamber B / m s^{-1}	acceleration in chamber B / m s^{-2}
A	12	2.5
B	24	2.5
C	48	5.0
D	64	5.0

16. [NJC/Prelims 2014/P1/13]

A race track has the shape of an inverted cone, as the drawing shows. On this surface the cars race in circles that are parallel to the ground.

For a speed of 34 m s^{-1} , at what value of the distance d should a driver locate his car if he wishes to stay on a circular path without depending on friction?



A 95 m

B 183 m

C 230 m

D 304 m

17. [NYJC/Prelims 2014/P1/11]

A particle of mass m moves in a circle of radius r at a uniform speed with frequency f . What is the kinetic energy of the particle?

A $\frac{mf^2 r^2}{4\pi^2}$

B $\frac{mf^2 r}{2}$

C $2\pi^2 m f^2 r^2$

D $4\pi^2 m f^2 r^2$

18. [NYJC/Prelims 2014/P1/12]

A car of mass m moving at a constant speed v passes over a humpback bridge of radius of curvature r . (A humpback bridge is curved in a semicircle above a river).



The car remains in contact with the road.

What is the net force exerted by the car on the road when it is at the top of the bridge?

A $mg + \frac{mv^2}{r}$

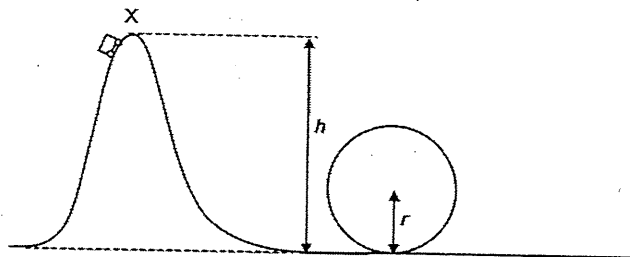
B $\frac{mv^2}{r}$

C mg

D $\frac{mv^2}{r} - mg$

19. [PJC/Prelims 2014/P1/11]

A roller coaster consists of a steep curve of height h and a circular loop of radius r .



The ride is designed such that once a car of mass m is brought up to the highest point X, and released, no further energy needs to be supplied to it to complete the loop safely. The total amount of work done against resistive forces from point X to the top of the loop is W .

What is the minimum height h in terms of m , r , W and g , the acceleration of free fall?

- A $\frac{5}{2}r + \frac{W}{mg}$ B $\frac{5}{2}r - \frac{W}{mg}$ C $\frac{3}{2}r + \frac{W}{mg}$ D $\frac{3}{2}r - \frac{W}{mg}$

20. [PJC/Prelims 2014/P1/12]

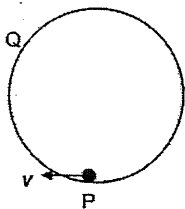
A car is moving at a constant speed of 5.5 m s^{-1} and is making a turn of radius 12 m. A small pendulum bob of mass 20 g hanging on the rear view mirror inside the car is observed to tilt at an angle θ to the vertical.

What is the value of angle θ ?

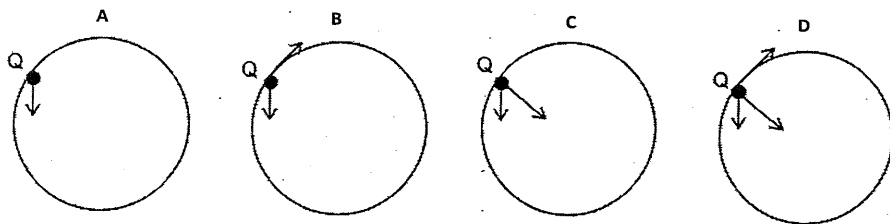
- A 2.7° B 14° C 15° D 75°

21. [RVHS/Prelims 2014/P1/10]

A small ball bearing is projected with a velocity v from the lowest position P of a vertical circular track which is not smooth. The ball bearing starts to leave the track at Q.

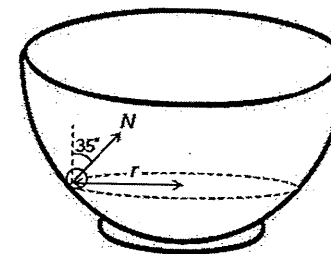


Which of the following represents all the forces acting on the ball-bearing at Q?



22. [SAJC/Prelims 2014/P1/11]

A small glass marble is moving in a horizontal circle round the inside surface of a smooth bowl. Its angular velocity is 8.0 rad s^{-1} and the normal reaction N acting on the marble inclined at 35° to the vertical as shown.



What is the radius r of the horizontal circle?

- A 0.050 m B 0.070 m C 0.090 m D 0.11 m

23. [SAJC/Prelims 2014/P1/12]

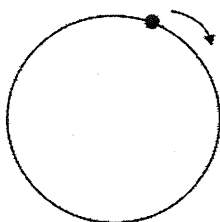
A man travelling eastward on a fast-moving train along the earth's equator has a weight W_e . Another man traveling westward on a fast-moving train along the Earth's equator has a weight W_w .

Which statement is correct?

- A $W_e > W_w$. The earth's linear speed at the equator due to its rotation is added to the speed of the east-bound train.
- B $W_e > W_w$. The speed of the east-bound train is subtracted from the earth's linear speed at the equator due to its rotation.
- C $W_w > W_e$. The speed of the west-bound train is subtracted from the earth's linear speed at the equator due to its motion.
- D $W_w > W_e$. The speed of the west-bound train is added to the earth's linear speed at the equator due to its motion.

24. [SRJC/Prelims 2014/P1/11]

A body moves in a uniform circular motion as shown.

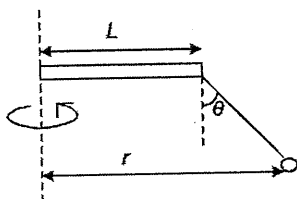


Which set of description correctly gives (i) the work done by the centripetal force, (ii) the change in the angular velocity of the body and (iii) the change in the tangential velocity when it is moving in this circular motion?

	work done by the centripetal force	angular velocity	tangential velocity
A	not zero	no change	change
B	zero	no change	no change
C	zero	no change	change
D	zero	change	no change

25. [SRJC/Prelims 2014/P1/12]

A pendulum bob is tied to an inelastic string which is in turn tied to one end of a horizontal bar of length L . The bar is set to rotate about a vertical axis at the other end with a period T . The string is found to make an angle θ to the vertical and the pendulum is at a distance r from the axis.

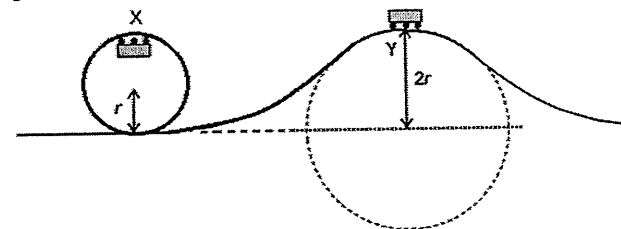


What is the value of $\tan \theta$?

- A $\frac{2\pi L}{Tg}$ B $\frac{2\pi r}{Tg}$ C $\frac{4\pi^2 r}{T^2 g}$ D $\frac{4\pi^2 (r-L)}{T^2 g}$

26. [TJC/Prelims 2014/P1/12]

In a roller coaster ride, a cart of mass m loops a loop of radius r before racing over a hump of radius of curvature $2r$ as shown in the diagram.

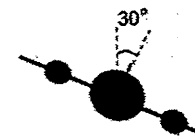


If the cart just remains in contact with the track at the top of the loop at point X, what is the force the cart exerts on the track at the top of the hump at point Y? Assume that resistive forces are negligible, and ignore the dimensions of the cart.

- A 0 B $\frac{mg}{2}$ C mg D $\frac{3mg}{2}$

27. [TJC/Prelims 2014/P1/13]

An aircraft is moving in a horizontal plane at a constant speed. It banks at angle of 30° to the vertical in order to make a turn as shown in the diagram below. The radius of the turn is 5000 m.

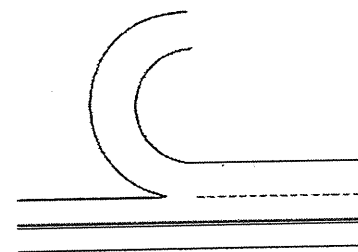


What is the speed of the aircraft?

- A 40 m s^{-1} B 54 m s^{-1} C 168 m s^{-1} D 291 m s^{-1}

28. [TPJC/Prelims 2014/P1/10]

Many highway entrance and exit roads are circular and designed for cars moving at 50 km h^{-1} .

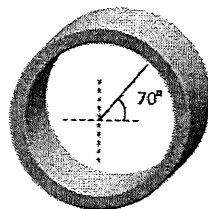


To design a similar road for a speed of 100 km h^{-1} , the engineers should increase the radius of the circular section by a factor of

- A $\sqrt{2}$ B 2 C 3 D 4

29. [TPJC/Prelims 2014/P1/11]

In an automatic clothes dryer, a hollow cylinder moves the clothes on vertical circle (radius $r = 0.32$ m) as shown in the figure below. The appliance is designed so that the clothes tumble gently as they dry.



This means that when a piece of clothing reaches an angle of 70° above the horizontal, it loses the contact with the wall of the cylinder and falls onto the clothes below.

What is the frequency of the rotating cylinder in order that the clothes lose contact with the wall at that angle?

- A 0.52 Hz B 0.85 Hz C 0.88 Hz D 5.54 Hz

30. [VIC/Prelims 2014/P1/11]

A car of mass 200 kg goes over a semicircular hill of radius 12 m at a speed of 8.0 m s^{-1} .

What is the normal reaction that it experiences from the road surface when it is at the top of the hill?

- A 0 N B 900 N C 2000 N D 3000 N

31. [VIC/Prelims 2014/P1/12]

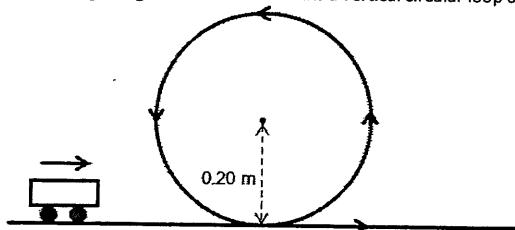
Muthu is playing with a marble in a bowl. The bowl is hemispherical with a radius of 5.0 cm, and has a smooth interior. He shakes the bowl horizontally, so that the marble performs horizontal circular motion at a height of 2.0 cm above the bottom of the bowl.

What is the period of the marble?

- A 0.35 s B 0.39 s C 0.52 s D 3.9 s

32. [YJC/Prelims 2014/P1/11]

A toy car of mass 0.50 kg is travelling along a track which contains a vertical circular loop of radius 0.20 m as shown.

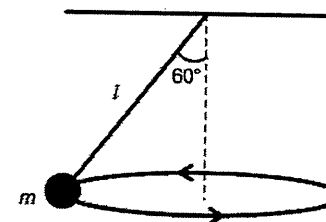


What is the minimum speed that the car must have when entering at the bottom of the track in order to prevent it from falling off at the top of the track?

- A 1.40 m s^{-1} B 2.80 m s^{-1} C 3.13 m s^{-1} D 6.20 m s^{-1}

33. [YJC/Prelims 2014/P1/12]

A pendulum bob of mass, m is tied to a string of length L . The pendulum bob is set to swing in a horizontal circle. The string makes an angle of 60° with the vertical, as shown.



Which of the following gives the correct expression for the tension in the string and the linear speed of the bob?

	tension	linear Speed
A	mg	$\sqrt{Lg}$
B	mg	$1.22\sqrt{Lg}$
C	$2mg$	$\sqrt{Lg}$
D	$2mg$	$1.22\sqrt{Lg}$

II. Structured Questions

1. [ACJC/Prelims 2014/P3/2]

- a. Explain what is meant by the radian. [1]
- b. Fig. 1.1 shows a stone of mass 150 g at point P which is at the top of a circular path. The stone is suspended on one end of a string. The radius of the circle is 70 cm and the speed of the stone at point P is 3.5 m s^{-1} .

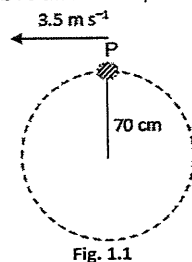


Fig. 1.1

- i. When the stone is at point P, calculate the resultant force acting on the stone. [2]
- ii. Hence, determine the tension in the string. [1]
- iii. On Fig. 2.1, draw and label the forces acting on the stone. [1]
- b. The speed of the stone remains constant at 3.5 m s^{-1} during the circular motion. On Fig. 2.2, sketch a force-time graph showing the variation of tension in the string as the stone travels a complete circle. [Take time = 0 s when stone is at point P] [2]
- Label M on your sketch the point of maximum tension in the string.

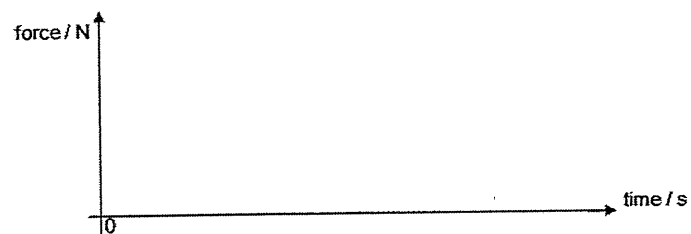


Fig. 1.2

[Total: 7]

2. [NJC/Prelims 2014/P2/4]

In the bagatelle board shown in Fig. 2.1, energy stored in a compressed spring is released to project a ball of mass 50 g along the surface of the board, which has a raised end and is inclined at 20° to the horizontal.

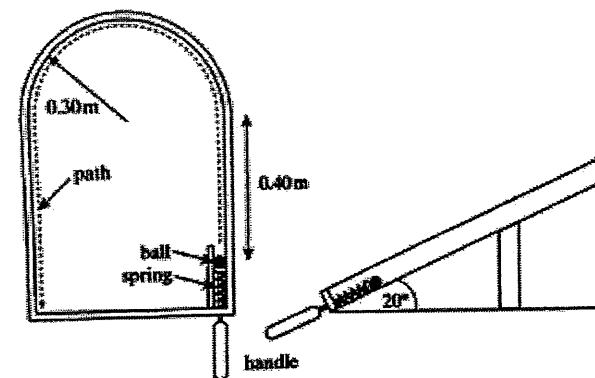


Fig. 2.1

The diagram shows the ball resting on the relaxed spring. Pulling on the handle with a force of 6.15 N compresses the spring by 50 mm and when the handle is released, the ball follows a path marked by the dotted line. Assume that frictional losses and the kinetic energy of the handle and spring are negligible. Ignore the effect of rotation.

- a. i. Calculate the speed of the ball as it reaches the top of the curve section. [2]
- ii. Draw a free body diagram showing all the forces acting on the ball at the instance in a. i.. [2]
- b. Show that the speed calculated in part a. i. is just sufficient for the ball to complete the path shown, explaining your reasoning. [3]

[Total: 8]

3. [SRJC/Prelims 2014/P3/1]

A horizontal flat plate is free to rotate about a vertical axis through its centre, as shown in Fig 3.1.

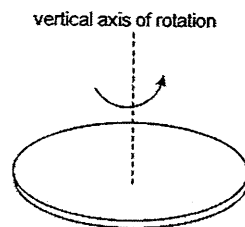


Fig. 3.1

A small object is placed on the plate, a distance of 0.35 m from the axis of rotation. The speed of rotation of the plate is gradually increased from zero until the mass is seen to slide off the plate.

The maximum frictional force F between the plate and the mass is given by the expression

$$F = 0.72W,$$

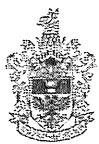
where W is the weight of the object.

- a. Determine the maximum number of revolutions of the plate per minute for the object to remain on the plate. [3]

- b. The plate in a. is covered, when stationary, with mud.

Suggest and explain whether mud near the edge of the plate or near the centre will first leave the plate as the angular speed of the plate is slowly increased. [2]

[Total: 5]



III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
D	B	A	C	C	B	C	C	C	C	D	B	D	B	B

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
B	C	D	A	B	A	D	C	C	C	B	C	D	B	B

31	32	33
A	C	D

- D Since $F_c = \frac{mv^2}{r}$, the force will be 4 times.
- B
 - A Correct. Using "right hand grip", the direction of angular velocity is perpendicular to the plane of circular motion.
 - B Incorrect. The work done by the centripetal force is zero since centripetal force (i.e. resultant force) acting on the mass is always perpendicular to the displacement travelled by the mass.
 - C Correct. When the object is back to the same position in one complete revolution, the velocity is the same. Hence, change in linear momentum of the body is zero.
 - D Correct. The total energy (i.e. KE & GPE) of the body remains constant.
- A Using $v = r\omega$, since angular velocity ω remains constant, hence speed v varies proportionally with r , which decreases at a constant rate.
- C

Since: $\omega = \frac{2\pi}{T}$ and $a_c = r\omega^2 = r\left(\frac{2\pi}{T}\right)^2$

$$\frac{a_m}{a_h} = \frac{r_m / T_m^2}{r_h / T_h^2} = \frac{4.3 / 3600^2}{2.7 / (3600 \times 12)^2} = 230$$

- C

At top, $N + mg = \frac{mv^2}{r}$, since $N \approx 0$, $\therefore v_{\text{top}}^2 = rg$

By conservation of energy,

$$\frac{1}{2}mv_{\text{top}}^2 + mgh = \frac{1}{2}mv_{\text{bottom}}^2$$

$$\frac{1}{2}rg + g(2r) = \frac{1}{2}v_{\text{bottom}}^2$$

$$5rg = v_{\text{bottom}}^2$$

$$v_{\text{bottom}} = \sqrt{(5)(1)(9.81)} = 7.00 \text{ m s}^{-1}$$
- B

Let radius of circular path be r . $\therefore r = 2 \sin \theta$

$$T \sin \theta = \frac{mv^2}{r} \Rightarrow T \sin \theta = \frac{mv^2}{2 \sin \theta} \Rightarrow \sin^2 \theta = \frac{mv^2}{2T}$$

$$T \cos \theta = mg \Rightarrow T^2 \cos^2 \theta = (mg)^2 \Rightarrow \cos^2 \theta = \frac{(mg)^2}{T^2}$$

$$\cos^2 \theta + \sin^2 \theta = \frac{(mg)^2}{T^2} + \frac{mv^2}{2T}$$

$$1 = \frac{(mg)^2}{T^2} + \frac{mv^2}{2T}$$

$$T^2 = (mg)^2 + \frac{mv^2 T}{2}$$

$$T^2 - 108T - (14.715)^2 = 0$$

$$T = 109.97 \text{ N} \quad \text{or} \quad -1.97 \text{ N (n.a.)}$$
- C

As the speed of the sphere entering the small loop is the same as the speed leaving the larger loop, hence

$$V = r_{\text{big}} \omega_{\text{big}} = r_{\text{small}} \omega_{\text{small}}$$

$$(16.2 \times 10^{-2})(63.0) = (9.7 \times 10^{-2}) \omega_{\text{small}}$$

$$\omega_{\text{small}} = 105.216 = 105 \text{ rad s}^{-1}$$
- C

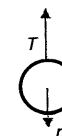
Considering forces when the string is vertical

$$T - mg = \frac{mv^2}{r} \Rightarrow T = \frac{mv^2}{r} + mg = \frac{(2.5)(0.475)^2}{0.8} + (2.5)(9.81) = 25.2 \text{ N}$$
- C

The car is speeding up at that position and therefore it has a tangential component of the acceleration (pointing downward). At the same time, it is traveling in a circular path so it has a radial component of the acceleration pointing towards the centre of the circle. So the net force which is the sum of these two components is best represented by C.
- C

$a_c = \omega^2 r = (2\pi f)^2 r_c = \left(2\pi \times \frac{1800}{60}\right)^2 0.36 = 12.8 \times 10^3 \text{ m s}^{-2}$

Centripetal force is acting towards the centre of the barrel by the wall.



11 D $F_d = 2F_w$

$$\frac{mv_d^2}{r} = 2 \frac{mv_w^2}{r}$$

Since $v_d = 16 \text{ m s}^{-1}$, $v_w = \sqrt{\frac{16^2}{2}} = 11 \text{ m s}^{-1}$

12 B $v = r\omega$, where ω is the angular velocity. $\Rightarrow \omega = \frac{v}{r}$

Since v is constant and r is decreasing, ω is increasing.

13 D Object is moving in circular motion, hence there must be a resultant force which is known as the centripetal force. The two forces acting are tension and weight.

14 B $r = \frac{343}{2\pi} = 54.6 \text{ m}$, $v = \frac{343}{22} = 15.6 \text{ m s}^{-1}$

$$F \cos \theta = mg, \quad F \sin \theta = \frac{mv^2}{r}$$

$$\tan \theta = \frac{v^2}{rg} = \frac{15.6^2}{54.6(9.81)} = 0.454 \Rightarrow \theta = 24.4^\circ$$

15 B $\omega = \frac{2\pi}{60} = \text{constant}$,

Centripetal acceleration of chamber A, $a_A = r_A \omega^2$. Similarly for chamber B, $a_B = r_B \omega^2$.

$$\frac{a_A}{a_B} = \frac{r_A}{r_B} = 4.0$$

$$\Rightarrow a_B = \frac{r_B}{r_A} \cdot a_A = \left(\frac{1}{4}\right)(10) = 2.5 \text{ m s}^{-2}$$

$$\therefore v_B = \frac{a_B}{\omega} = \frac{2.5}{(2\pi/60)} = 24 \text{ m s}^{-1}$$

16 B Radius of circular path = $d \cos 40^\circ$

$$N \cos 40^\circ = mg, \quad N \sin 40^\circ = \frac{mv^2}{r}$$

$$\frac{\sin 40^\circ}{\cos 40^\circ} = \frac{v^2}{rg} \Rightarrow r = \frac{v^2 \cos 40^\circ}{g \sin 40^\circ} \Rightarrow d \cos 40^\circ = \frac{v^2 \cos 40^\circ}{g \sin 40^\circ}$$

$$\therefore d = \frac{v^2}{g \sin 40^\circ} = \frac{(34)^2}{(9.81)(\sin 40^\circ)} = 183.3 \text{ m}$$

17 C Since speed is uniform, m , r and f remain constant.

$$\text{KE} = \frac{1}{2}mv^2 = \frac{1}{2}m(r\omega)^2 = \frac{1}{2}mr^2(2\pi f)^2 = 2mr^2\pi^2 f^2$$

18 D For car to remain in circular motion,

$$mg - N = \frac{mv^2}{r} \Rightarrow N = mg - \frac{mv^2}{r} \quad (\text{force of road on car})$$

By Newton's third law, $N_{\text{car on road}} = -N_{\text{road on car}} = -(mg - \frac{mv^2}{r}) = \frac{mv^2}{r} - mg$

19 A At top of loop, for the ride to continue without additional energy safely, $\frac{mv^2}{r} \geq mg$

Hence, $\frac{1}{2}mv^2 \geq \frac{1}{2}mgr$

Initial gravitational potential energy = Final total energy + work done against resistive forces

$$mgh = \frac{1}{2}mv^2 + mg(2r) + W$$

$$mgh \geq \frac{1}{2}mgr + mg(2r) + W$$

$$mgh \geq \frac{5}{2}mgr + W$$

$$h \geq \frac{5}{2}r + \frac{W}{mg}$$

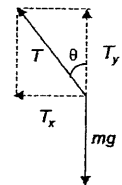
20 B The bob is also moving at a radius of 12 m with a speed of 5.5 m s⁻¹.

$$T_x = T \sin \theta = \frac{mv^2}{r}$$

$$T_y = T \cos \theta = mg$$

$$\tan \theta = \frac{v^2}{rg} = \frac{(5.5)^2}{(12)(9.81)} = 0.257$$

$$\theta = 14.4^\circ$$



21 A Leaves the track, hence no contact force and friction. Only force acting on the object is weight.

22 D $N \sin 35^\circ = mr\omega^2$, $N \cos 35^\circ = mg$

$$\tan 35^\circ = \frac{r\omega^2}{g}$$

$$\therefore r = \frac{g \tan 35^\circ}{\omega^2} = \frac{(9.81)(\tan 35^\circ)}{8^2} = 0.107 \text{ m}$$

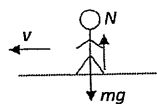
- 23 C Earth rotates West
- $\rightarrow$
- East

From an external observer:

Eastbound train linear speed > Westbound train linear speed.

Weight on the equator (the apparent weight on a scale, i.e. N)

Man undergoes circular motion about earth:

actual weight (W) $- N = \frac{mv^2}{r}$. Hence N will depend on the linear velocity of the train. $\therefore N = W_{\text{actual}} - \frac{mv^2}{r}$. Hence when $v \uparrow$, $N \downarrow$. Since $v_E > v_W$, hence $N_W > N_E$ Note: the actual weight is mg (constant). Weight of the man in question is with reference to his apparent weight, i.e. what is reflected on a weighing scale should a man be standing on it.

- 24 C Since the centripetal force is perpendicular to the direction of the tangential velocity, there is zero work done by the centripetal force.

There is no change in the angular velocity since there is no change in the direction (clockwise) and no change in its magnitude ($\omega = v/r$ and tangential speed is constant due to uniform circular motion)

The tangential velocity is changing since the direction is changing while the magnitude is kept constant.

25 C $\tan \theta = \frac{mr\omega^2}{mg} = \frac{r\omega^2}{g}$. Since $T = \frac{2\pi}{\omega}$, $\tan \theta = \frac{r(2\pi)^2}{T^2g} = \frac{4\pi r^2}{T^2g}$

26 B At X, $N_x + mg = \frac{mv^2}{r} \Rightarrow 0 + mg = \frac{mv^2}{r} \Rightarrow v = \sqrt{rg}$

Since GPE is the same, KE at X = KE at Y.

At Y, $mg - N_y = \frac{mv^2}{2r} \Rightarrow N_y = mg - \frac{m(rg)}{2r} \Rightarrow N_y = \frac{mg}{2}$

27 C $L \cos \theta = mg$ --- (1)

$L \sin \theta = mv^2/r$ --- (2)

(2)/(1) $\tan \theta = v^2/rg$

$v = \sqrt{5.0 \times 10^3 (9.81) \tan 30^\circ} = 168 \text{ m s}^{-1}$

- 28 D Friction provides for centripetal force, and friction remains constant.

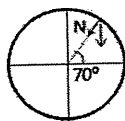
$F_c = \frac{mv^2}{r}$. If velocity increases by 2 times, then for the same F_c then r must increase by a factor of 4.

- 29 B Net resultant force at that point = component of weight in the radial direction +
- N
-
- For object to just lose contact, Normal reaction force
- $\rightarrow 0$
- .

$F_{\text{net}} = mg \sin 70^\circ + N = mr\omega^2$

$mg \sin 70^\circ = mr(2\pi f)^2$

$f = 0.8542 \text{ Hz}$



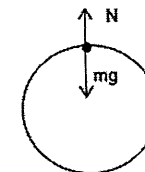
- 30 B

$mg - N = \frac{mv^2}{r}$

$N = mg - \frac{mv^2}{r}$

$N = (200)(9.81) - \frac{(200)(8.0)^2}{12}$

$N = 895 \text{ N}$



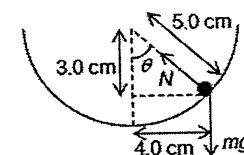
- 31 A Horizontally,
- $N \sin \theta = \frac{mr\omega^2}{r}$
- (1)

Vertically, $N \cos \theta = mg$ (2)

(1)/(2) $\tan \theta = \omega^2 r / g$

$\omega = 2\pi/T = \sqrt{\frac{g \tan \theta}{r}}$

$T = 2\pi \sqrt{\frac{r}{g \tan \theta}} = 2\pi \sqrt{\frac{0.040}{9.81 \times \frac{4}{3}}} = 0.35 \text{ s}$



- 32 C Refer to question 5. Similar method.

- 33 D

$T \sin 60^\circ = \frac{mv^2}{r}$ --- (1)

$T \cos 60^\circ = mg$ --- (2)

(1)/(2): $\tan 60^\circ = \frac{v^2}{rg}$ and $r = l \sin 60^\circ$

$v^2 = rg \tan 60^\circ = (l \sin 60^\circ) g \tan 60^\circ$

$v = 1.22\sqrt{lg}$

From (2): $T = \frac{mg}{\cos 60^\circ} = 2mg$

Structured Questions

1. [ACJC/Prelims 2014/P3/2]

a. See definitions in lecture notes.

b. i. Resultant force downwards provides centripetal force

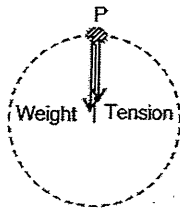
$$F_R = \frac{mv^2}{r} = \frac{(0.15)(3.5)^2}{0.7} = 2.63 \text{ N}$$

ii. $T + W = 2.63 \text{ N}$

$$T = 2.63 - W$$

$$T = 2.63 - 1.4715 = 1.1585 = 1.16 \text{ N}$$

iii.

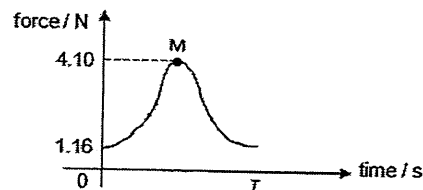


c. At the bottom of its path

$$\text{Tension} = 2.63 + W$$

$$\text{Tension} = 2.63 + 1.4715 = 4.096 = 4.10 \text{ N}$$

$$\text{Point M} = 4.10 \text{ N}$$



2. [NJC/Prelims 2014/P2/4]

a. i. Applying the Principle of Conservation of Energy on the energy of the ball:

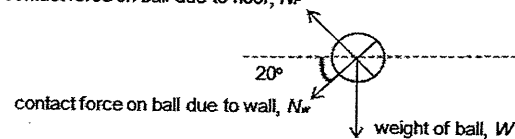
Loss in elastic potential energy = Gain in kinetic energy + gain in gravitational potential energy

$$\frac{1}{2}Fx = \frac{1}{2}mv^2 + mgh$$

$$\frac{1}{2}(6.15)(50 \times 10^{-3}) = \frac{1}{2}(0.050)v^2 + (0.050)(9.81)(50 \times 10^{-3} + 0.70) \sin 20^\circ$$

$$v = 1.06 \text{ m s}^{-1}$$

ii. contact force on ball due to floor, N_F



b. For the ball to complete the circular motion, the centripetal force is provided by the contact force on ball due to wall and component of the weight that is directed to the centre of the circle.

Using Newton's 2nd law: $F_{\text{net}} = ma$

$$N_w + W \sin 20^\circ = mv^2/r$$

$$N_w = mv^2/r - W \sin 20^\circ$$

For circular motion to be completed, $N_w > 0$

$$mv^2/r - W \sin 20^\circ > 0$$

$$mv^2/r > mg \sin 20^\circ$$

$$v > 1.003 \text{ m s}^{-1}$$

Since the speed in a. i. is greater than the minimum speed required, the ball is able to complete the circular path.

3. [SRJC/Prelims 2014/P3/1]

a. Friction between mass and plate provides the centripetal force.

$$0.72 W = mr\omega^2$$

$$0.72 m(9.81) = m(0.35)\omega^2$$

$$\omega = 4.49 \text{ rad s}^{-1}$$

$$n = \frac{\omega}{2\pi}(60) = 43 \text{ revolutions per min}$$

b. At edge of the plate, the value of r is larger and hence (by $F = mr\omega^2$) it would require a larger centripetal force acting on the mud to hold it in position.

(In this case, it is the friction between the mud and plate that provides the centripetal force.) Hence, as the angular speed is slowly increased, the mud near the edge, which requires a higher centripetal force, will have the required centripetal force exceeding the maximum frictional force first and hence have a higher tendency of flying off the wheel.